Jake Amaral

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Education

University of Florida Aug 2024 – May 2028

*B.S. in **Computer Science** and Minor in **Economics*** *Gainesville, FL*

- **GPA:** 3.52
- **Relevant Coursework:** Data Structures & Algorithms, Prog in C++, Prog in Python, Linear Algebra
- **Extracurricular:** Kappa Sigma Fraternity, Association for Computing Machinery

Technical Skills

Languages: Python, C++, C#, SQL (PostgreSQL), HTML, TypeScript
Frameworks: Node.js, Next.js, React, Tailwind, shadcn, Drizzle ORM, AWS SAM
Developer Tools: Git/GitHub, Vercel, Supabase, Clerk, Unity, Google OAuth, AWS, Docker, React Native

Experience

Design Team – Product Manager Sep 2026 – Present

UF Association for Computing Machinery *Gainesville, FL*

- Developing an event check-in platform for **UF's largest pre-professional computing org (2,300+ members)**, scaling to **thousands** of students across partner orgs (Next.js, TypeScript, Supabase/PostgreSQL)
- Building a cross-platform iOS/Android app with Expo (React Native) on a shared Supabase backend secured by Clerk JWT auth and row-level security
- Leading mobile team onboarding as PM: scope Jira tickets, assign work by developer strengths, and guide each member through shipping a PR via code review and CI

Software Engineering Intern May 2026 – Jul 2026

Safe Cities NPO *Cape Town, South Africa*

- Led a 4-person team to build and ship a full-stack platform now used by **50+** nonprofit staff, using Next.js, TypeScript, SQL (PostgreSQL), and Drizzle ORM
- Enforced role-based access for every user by implementing Clerk authentication with Google OAuth 2.0 and Calendar API integrations
- Delivered a production-ready application from ambiguous requirements by interviewing stakeholders and iterating designs directly with Safe Cities staff
- Gained a broader perspective on global software development and cross-cultural collaboration through **direct cultural immersion**

Projects

Spatial Math | *C#, Unity, Python, AWS (Bedrock, Lambda, SAM), Docker* Apr 2026 – Present

- Built a real-time 3D graphing calculator in Unity that compiles user-typed equations into executable functions, with variables, nested functions, operator precedence, and implicit multiplication
- Rendered continuous height-mapped surfaces with procedural mesh generation sampling **3,700+ vertices**, plus click-to-inspect raycasting that computes and renders a 3D tangent plane via numerical differentiation
- Deployed a **serverless AWS backend** (SAM, Lambda, Docker) pairing the deterministic C# engine with an LLM on Amazon Bedrock that explains results but never calculates them
- Cut costs and attack surface with DynamoDB response caching, rate-limited API keys, least-privilege IAM roles, and prompt-injection-hardened input validation

Enterprise Decision Making (EDM) AI Hackathon Entry | *IBM watsonx Orchestrate* Feb 2026

- Designed a workflow of **6 agents** in IBM's watsonx AI Agent Builder to support business decision-making and governance at scale
- Surfaced inconsistencies and policy concerns hidden across **60+** synthetic data points by building specialized extraction agents that parsed a structured JSON dataset
- Preserved human authority over final outcomes by architecting an auditable review process, improving decision consistency and transparency without removing the human from the loop

Languages & Interests

Languages: English (Native), Spanish (Intermediate High), Portuguese (Basic)
Hobbies: Baseball, Electronic Music Production, Vibe-Coding, Golf, Martial Arts, Traveling